



Höhere Technische Bundeslehranstalt Wels

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ELECTROTECHNICAL LABORATORY

Title: **10. Oscilloscope**

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Team members:

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Used instruments:

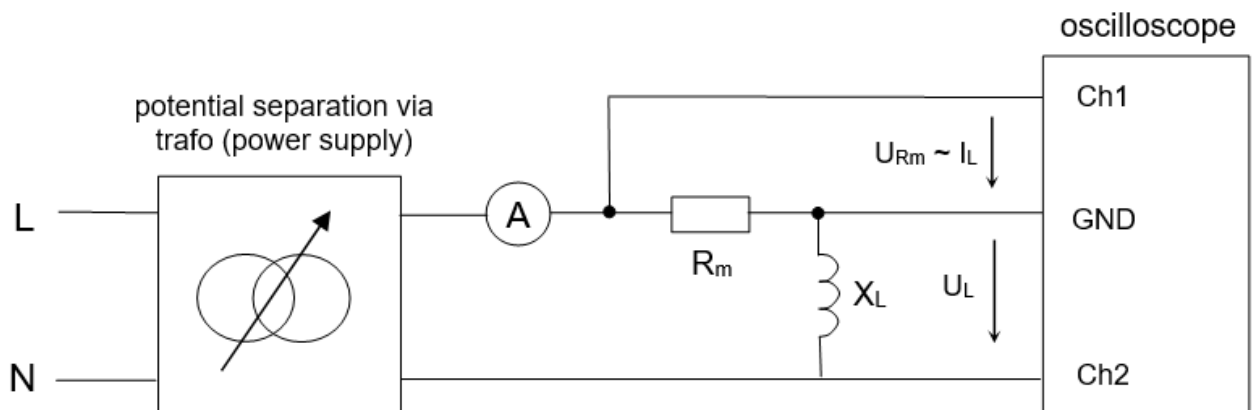
Gwinstek GDS-1022 824-30-229; W1.17-227 - oscilloscope
Gwinstek AFG-2109 824-30-327; W1.17-736 - isolation amp.
Gwinstek AFG-2109 824-30-327; W1.17-737 - isolation amp.
Fluke True RMS W1.17-52
10 Ω -slider resistor
100 Ω -slider resistor
coil with 600 windings + iron core

Remarks: Delayed handing in because of one week absence.

At the beginning of our first lesson with the oscilloscope we tried to make ourselves a quick overview of the used oscilloscope (Gwinstek GDS-1022 824-30-229) and tried the different functions by using a function generator (Gwinstek AFG-2109 824-30-327).

10.1 Phase angle of impedances:

Preparation:



oscilloscope: Gwinstek GDS-1022 824-30-229; W1.17-227

A-Meter: Fluke True RMS W1.17-52

R_m (R_{Shunt}): 10 Ω-slider resistor; I_{max} = 5.7A

X_L: coil with 600 windings; I_{max} = 2.5A

R_m = 10.6 Ω (max. of 10 Ω-slider resistor)

R_{XL} = 2.5 Ω

U_{max} = I_{max} * R_{ges} = 35.25V

For this task, we had to measure the phase angle and the voltage and current waveform of an impedance. We used a simple coil with and without an iron core for it. Because of the "ground-problem", not to short the circuit, by connecting the ground of the oscilloscope in between R_m and X_L had to use a power supply with a trafo for potential separation. Also the signal of the second channel was inverted, so we had to change it in the menu of the oscilloscope.

Next up we had to set both signals of the two channels to the same zero-line to manually measure the time difference of the zero-crossings using the cursor function of the oscilloscope.

By using the measuring function of the oscilloscope, the peak value of the signal can be shown.

There we noticed that the sin-curve of the voltage of the school's network was a bit deformed because of the big amount of consumers.

Measurement:

$$f = 50 \text{ Hz}$$

$$\varphi = \omega \cdot \Delta t$$

without iron core:

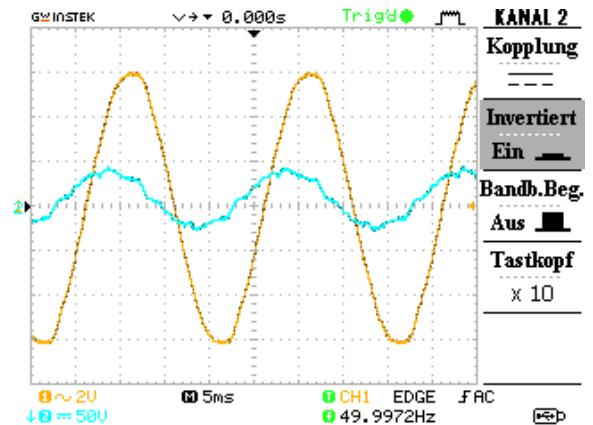
$$\Delta t = 3 \text{ ms} \rightarrow \varphi = 0.9425 \text{ rad} = 54^\circ$$

U = 6V AC (set at the power supply)

I = 0.403 A (A-Meter)

channel 1 = 5.92V peak

channel 2 = 35.2V peak



with iron core (top opened):

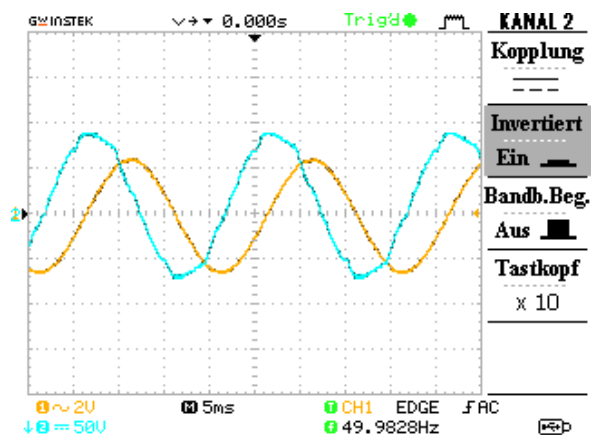
$$\Delta t = 4.6 \text{ ms} \rightarrow \varphi = 1.445 \text{ rad} = 82.8^\circ$$

U = 6V AC (set at the power supply)

I = 0.164 A (A-Meter)

channel 1 = 2.4V peak

channel 2 = 80.8V peak



with iron core (totally closed):

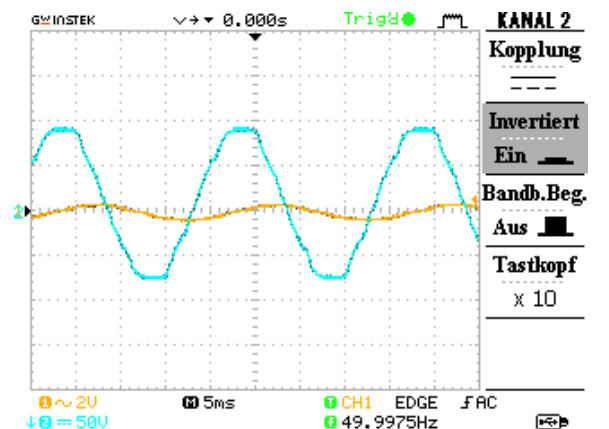
$$\Delta t = 4.4 \text{ ms} \rightarrow \varphi = 1.382 \text{ rad} = 79.2^\circ$$

U = 6V AC (set at the power supply)

I = 0.023 A (A-Meter)

channel 1 = 300mV peak

channel 2 = 85.5V peak



Conclusion:

It looks like the resistance of the coil goes up, when the iron core is being applied to it. Maybe because reduced eddy currents.

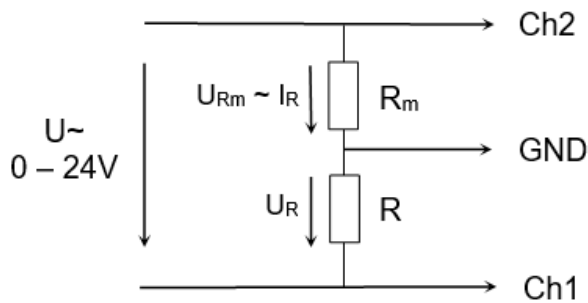
Also the inductance is closer to an ideal impedance when using an iron core because φ should be 90° .

The voltage and current on the graphs look very different because of the different zooms in each diagram.

10.2 Visualization of characteristics:

1) $R_1 \approx 10 \Omega$

Preparation:



oscilloscope: Gwinstek GDS-1022 824-30-229; W1.17 227

R_m (R_{shunt}): 10 Ω -slider resistor; $I_{\text{max}} = 5.7\text{A}$

R_1 : 100 Ω -slider resistor; $I_{\text{max}} = 1.8\text{A}$

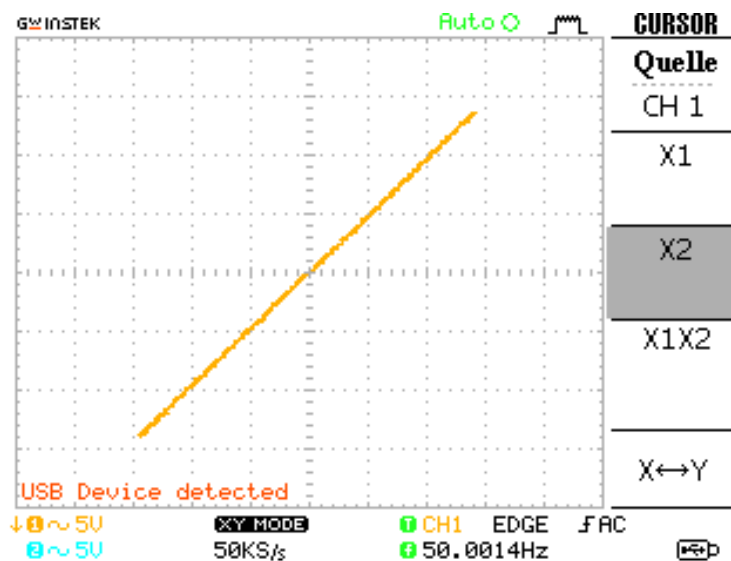
$R_m = 10.6 \Omega$ (max. of 10 Ω -slider resistor)

$R_1 = 9.7 \Omega$ (set 100 Ω -slider resistor with Fluke True RMS 177 W1.17-52)

$$U_{\text{max}} = I_{\text{max}} * R_{\text{ges}} = 36.54\text{V}$$

$$U_{\sim} = 20\text{V}$$

For this task my partner and I had to set up the oscilloscope once again, but this time to draw channel 1 over channel 2. This function is available in the "menu" and is marked with "XY". Then channel 1 becomes the x-axes and channel 2 becomes the y-axes.



Measurements:

channel1 = 28.6V peak
channel 2 = 27.6V peak

The reverse of the slope of the drawn figure now should be relation between the first and second resistor. ($R_1/R_m = ch1/ch2$) ($k=\Delta y/\Delta x=ch2/ch1$)

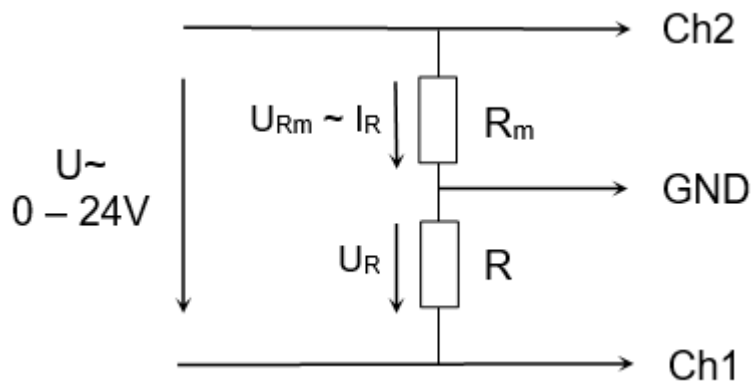
$$R = ch1/ch2 * R_m = 28.6/27.6 * 10.6 = 10.98 \Omega$$

Conclusion:

The calculation of the resistance is within the margin of error because of the difficulty measuring the peaks of the drawn graphs with the cursor function.

2) $R_1 \approx 50 \Omega$

Preparation:



oscilloscope: Gwinstek GDS-1022 824-30-229; W1.17 227

R_m (R_{shunt}): 10 Ω -slider resistor; $I_{max} = 5.7A$

R_2 : 100 Ω -slider resistor; $I_{max} = 1.8A$

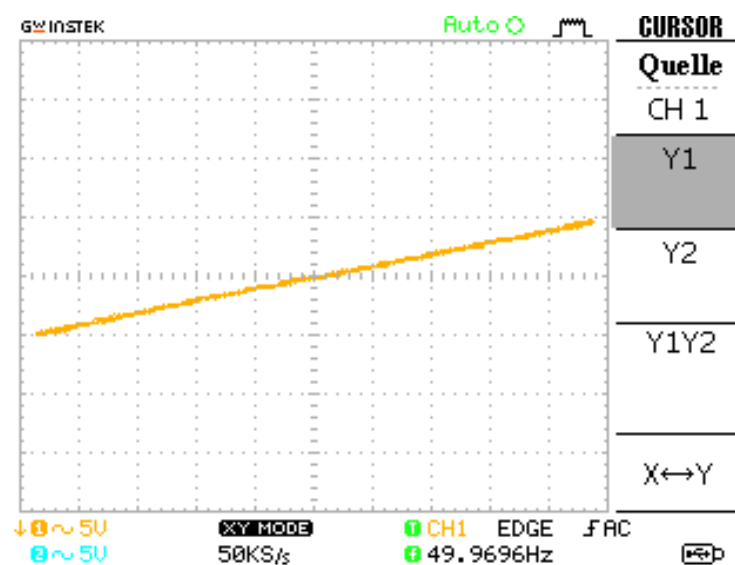
$R_m = 10.6 \Omega$ (max. of 10 Ω -slider resistor)

$R_2 = 50.3 \Omega$ (set 100 Ω -slider resistor with Fluke True RMS 177 W1.17-52)

$U_{max} = I_{max} * R_{ges} = 109.62V$

$U_{\sim} = 20V$

The oscilloscope was kept at the same settings, only the scale had to be modified to get the whole Lissajous-figure on the screen.



Measurements:

channel1 = 47.4V peak

channel 2 = 9.8V peak

The reverse of the slope of the drawn figure now should be relation between the first and second resistor. ($R_1/R_m = ch1/ch2$) ($k=\Delta y/\Delta x=ch2/ch1$)

$$R = ch1/ch2 * R_m = 47.4/9.8 * 10.6 = 51.27 \Omega$$

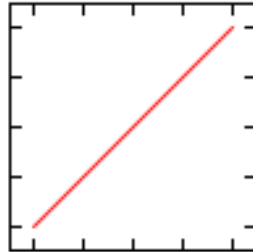
Conclusion:

Again the calculation isn't completely correct, but with an error of approximately 2% is okay.

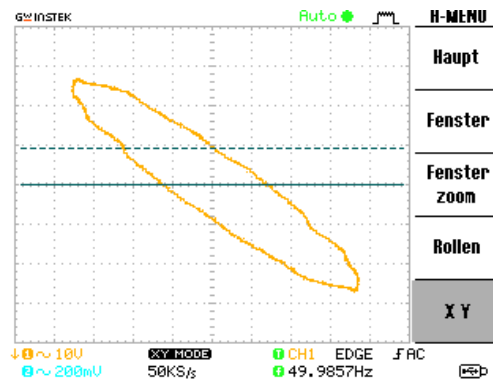
10.3 Lissajous-figures:

1) $f_1 = f_2 = 50 \text{ Hz}$

Soll: ($\varphi=0$)

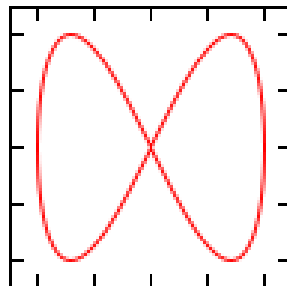


Ist:

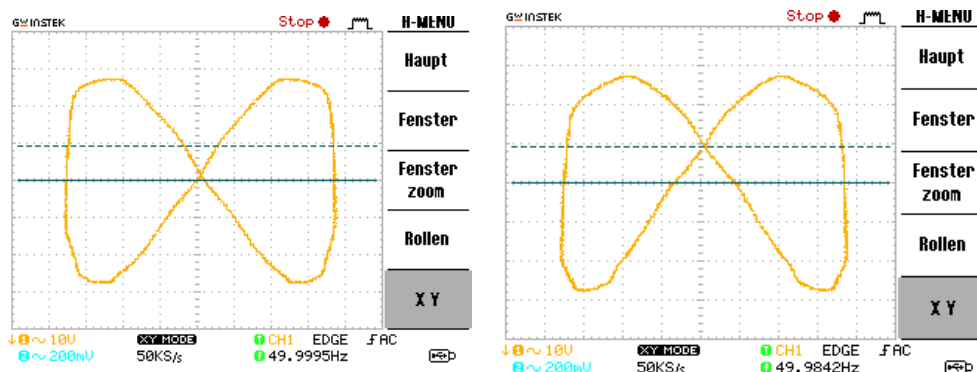


2) $f_1 = 50 \text{ Hz}; f_2 = 100 \text{ Hz}$

Soll: ($\varphi=0$)

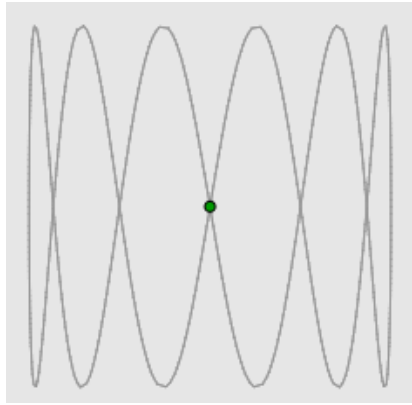


Ist:

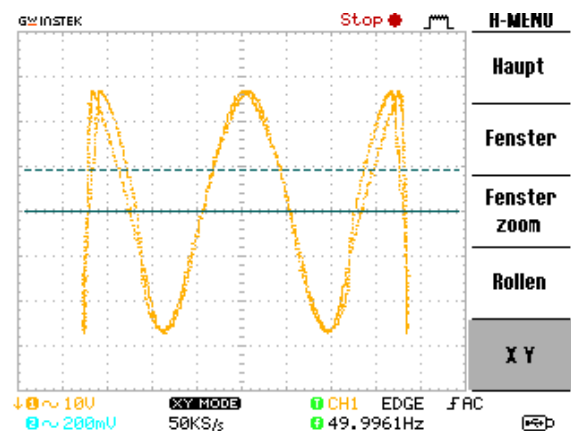
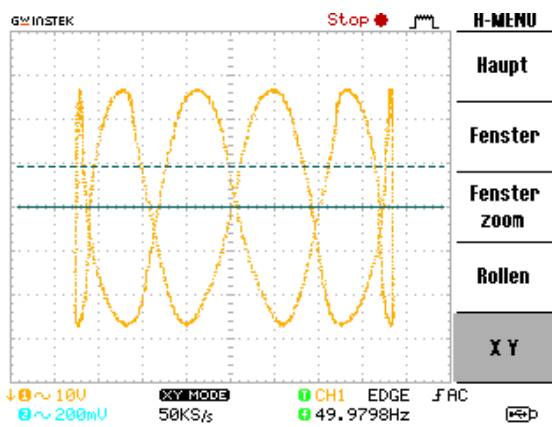


3) $f_1 = 50 \text{ Hz}$, $f_2 = 300 \text{ Hz}$

Soll: ($\varphi=0$)



Ist:



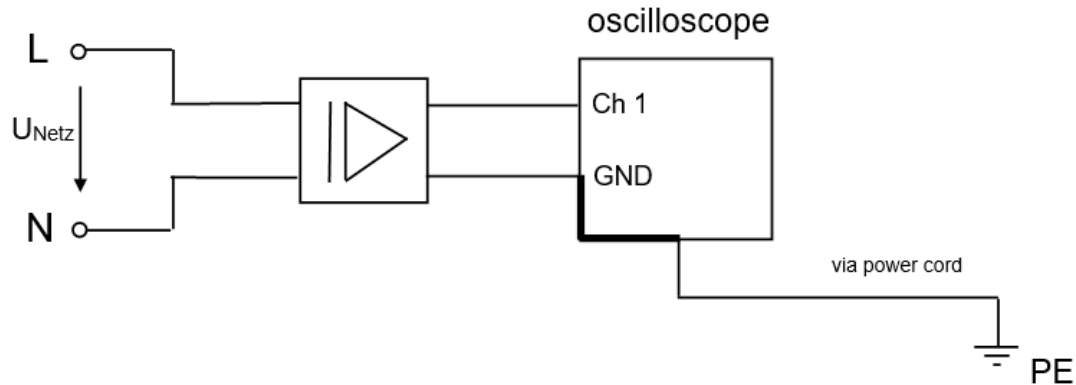
Conclusion:

Because the power supply and the function generator can't create perfect sin-curves with the exact frequency as needed and with sometimes a small phase angle, the figures may seem to "move" slowly. In our screenshots taken, the Lissajous-figure therefore might look a bit odd.

10.4 Measuring the net voltage – ground problem:

a) one phase:

Preparation:



oscilloscope: Gwinstek GDS-1022 824-30-229; W1.17-227

isolation amplifier: Gwinstek AFG-2109 824-30-327; W1.17-736

In this task we had to measure and draw the graph of the net voltage. Because of safety reasons we only used a power supply that outputted 23V alternating current. Because the oscilloscope connects the channels to a ground signal we needed put an isolation amplifier in between the power supply and oscilloscope not to create a short circuit.

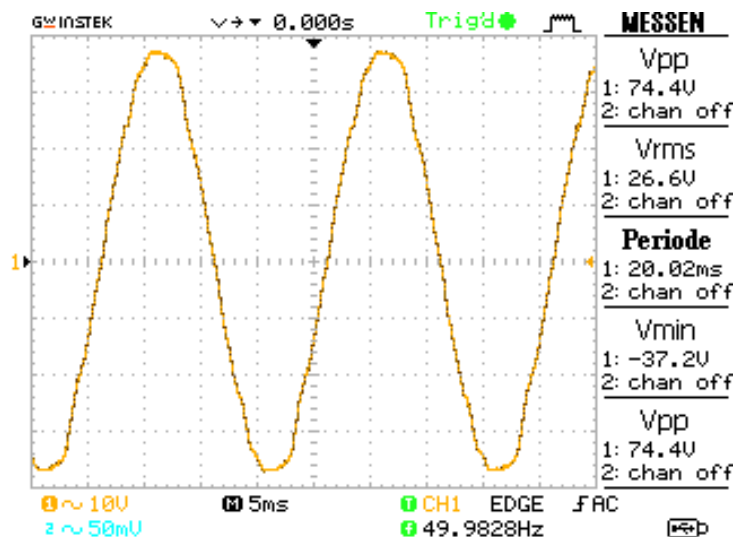
Measurement:

Ist:

$U_{\text{pp}} = 74.4 \text{ V}$
 $U_{\text{rms}} = 26.6 \text{ V}$
 $f = 49.98 \text{ Hz}$

Soll:

$U_{\text{pp}} = 65.05 \text{ V}$
 $U_{\text{rms}} = 23 \text{ V}$
 $f = 50 \text{ Hz}$



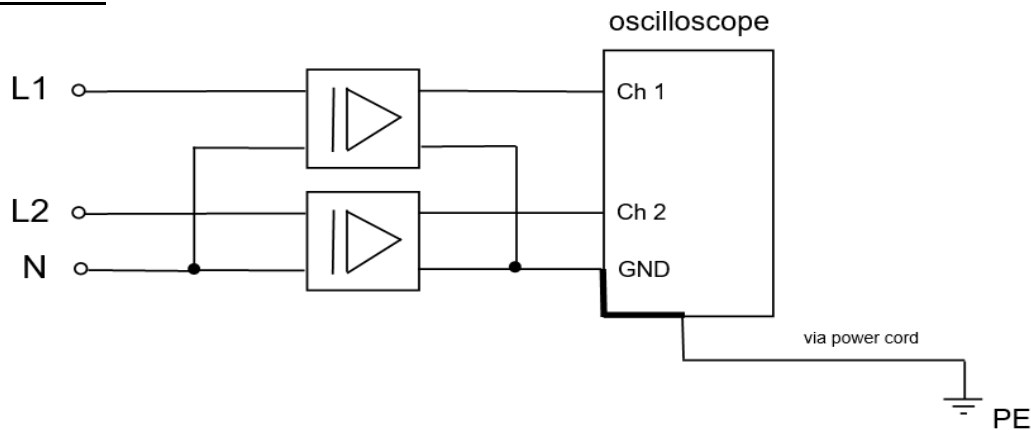
Conclusion:

The power supply doesn't put out exactly the 23 Volts alternating current as stated, but the frequency is almost exactly as expected.

You always should be careful, that the isolation amplifier still has some battery left.

b) two phases each

Preperation:



oscilloscope: Gwinstek GDS-1022 824-30-229; W1.17-227

isolation amplifier for L1: Gwinstek AFG-2109 824-30-327; W1.17-736

isolation amplifier for L2: Gwinstek AFG-2109 824-30-327; W1.17-737

In addition to the previous task we now had to measure two phases of the three phase power supply at the same time to also find out their phase shift. In addition we also had to measure the phase-to-phase voltage using the diff-function on the oscilloscope.

1) channel 1 ... L1; channel 2 ... L2

Measurements:

Ist:

- channel1:

$$U_{pp} = 73.6 \text{ V}$$

$$U_{rms} = 26.5 \text{ V}$$

$$f = 50.00 \text{ Hz}$$

- channel1:

(The oscilloscope only shows 1/10 because we still had the multiplier turned on)

$$U_{pp} = 71.6 \text{ V}$$

$$U_{rms} = 26.0 \text{ V}$$

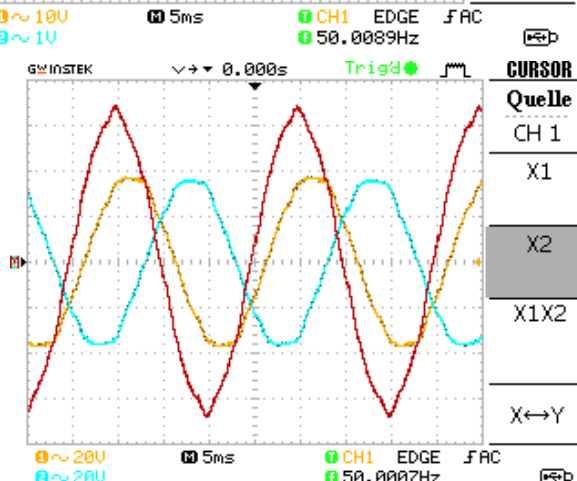
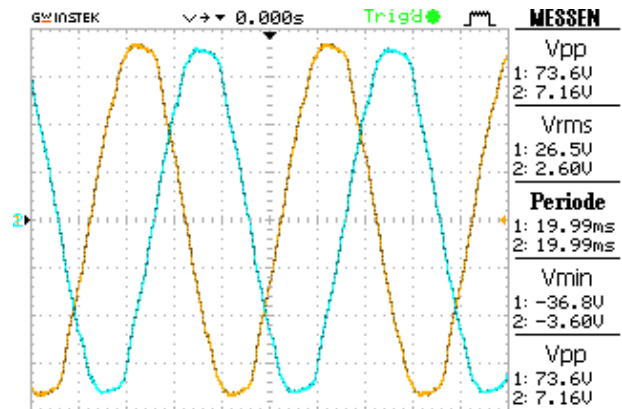
$$f = 50.00 \text{ Hz}$$

- ch1-ch2:

(diff-function; black)

$$U_{pp} = 135 \text{ V}$$

$$U_{rms} = U_{pp}/\sqrt{2} = 47.73 \text{ V}$$



$$\Delta t = 6.8 \text{ ms} \rightarrow \varphi = 2.14 \text{ rad} = 122.4^\circ$$

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Soll:

- channel1:

$$U_{pp} = 65.05 \text{ V}$$

$$U_{rms} = 23 \text{ V}$$

$$f = 50 \text{ Hz}$$

- channel1:

(The oscilloscope only shows
1/10 because we still had the multiplier
turned on)

$$U_{rms} = 65.05 \text{ V}$$

$$U_{rms} = 23 \text{ V}$$

$$f = 50 \text{ Hz}$$

- ch1-ch2:

(diff-function; black)

$$U_{pp} = 113.12 \text{ V}$$

$$U_{rms} = U_{pp}/(2\sqrt{2}) = 40 \text{ V}$$

$$\varphi = 2.094 \text{ rad} = 120^\circ \text{ (if correct phase sequence)}$$

2) channel 1 ... L1; channel 2 ... L3

Ist:

$$\Delta t = 13.8 \text{ ms} \rightarrow \varphi = 4.335 \text{ rad} = 248.4^\circ$$

Soll:

$$\varphi = 4.189 \text{ rad} = 240^\circ \text{ (if correct phase sequence)}$$

Conclusion:

As in the previous task, the used power supply doesn't put out the exact voltage, but the phase sequence is correct: L1 – L2 – L3, because the phase angle between each phase is approximately 120° (1/3 of 360°).